

Pre-Launch Procedures

Hobby Rocket Launch

Vehicle: Hobby rocket

Avionics firmware: v2.0.0 (MPR Altitude Logger)

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Conventions

T-N = N days before launch day. T-0 = launch day at the field.

GO / NO-GO blocks at the end of each phase are mandatory checkpoints — if any condition fails, do not proceed without the relevant lead's sign-off.

All `npm` commands run from `~/Documents/mpm-altitude-logger` on Dash's laptop.

Abbreviations

- **ORK** — OpenRocket simulation file
- **lwr** — lower (bulkhead)
- **AGL** — altitude above ground level
- **WDT** — watchdog timer (reboot if firmware hangs)
- **SD** — SD card (microSD on the avionics PCB)
- **TUI** — text-based UI on the laptop (preflight / postflight tools)
- **PCB** — printed circuit board (the avionics flight computer)

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Section 1

Aerostructures

Airframe build, ballast, integration

1.1 T-1

Check wind predictions for the day

- ☐ Pull forecast for the launch site, hour-by-hour for the launch window
- ☐ Conduct additional ORK simulations at predicted windspeed $\pm 20\%$
- ☐ Calculate appropriate ballast for windspeed at predicted $\pm 50\%$
- ☐ Record the ballast mass for each scenario in the team sheet

1.2 T-0 (Launch Day)

Install eyebolt in lower bulkhead

- ☐ Drill as close to the centre as possible
- ☐ Use the pre-marked drill jig if available
- ☐ Confirm eyebolt threads engage cleanly before epoxy step

Motor integration

- ☐ Friction-fit the motor into the airframe
- ☐ Use the motor retention clip to retain the motor
- ☐ Confirm clip is fully seated — no visible gap

Measure wind

- ☐ Use the handheld anemometer (the “fan-lookin’ thing”)
- ☐ Record three readings 30 seconds apart at pad height
- ☐ If wind exceeds the launch limit from your ORK sims, hold

Add ballast to the tip of the nose cone

- ☐ Mix up some 5-minute epoxy
- ☐ Log weight of mixed epoxy
- ☐ Measure split shot weight: $W_{ss} = W_{ballast} - W_{epoxy}$
- ☐ Place split shot sinkers at the tip of the nose cone, then add the epoxy
- ☐ Hold the nosecone nose-down until the epoxy cures (~5 min, verify with toothpick)

Add battery (coordinate with Avionics — battery only goes in AFTER avionics bench check passes)

- ☐ Connect battery to the flight computer
- ☐ Battery life on fresh 9V alkaline: ~4+ hours idle. Install within 1 hour of launch slot. Set a phone timer.
- ☐ Lower battery into the slot in the nose cone
- ☐ Secure battery: electrical tape preferred. Hot glue is a fallback only — accept that it may shift under boost loads.

Install flight computer

- ☐ Lower the connected PCB into the nosecone
- ☐ Use 3× M3 magnetic screws + screwdriver to screw the PCB in place
- ☐ DO NOT overtighten — PLA threads strip easily

- ☐ Verify slow-blink LED is still visible through the nosecone vent / pressure hole before sealing
- ☐ Note: the lower bulkhead glue (next step) seals the bay. Post-flight recovery requires cutting the nosecone open — the screws are only accessible after that cut.

Install lower bulkhead

- ☐ Mix up 5-minute epoxy
- ☐ Lather the sides and rear of the bulkhead with epoxy
- ☐ Using the eyebolt, hold the bulkhead in place until epoxy cures
- ☐ Tie the shock cord to the eyebolt

NOSECONE DONE!

Integrate nosecone with main airframe

Drill pressure hole

- ☐ Take care not to drill into the avionics bay or shock cord
- ☐ Hole diameter per design spec — confirm before drilling

GO / NO-GO

GO: ballast logged, motor retained, bulkhead cured, pressure hole drilled, avionics LED slow-blinking through pressure hole.

NO-GO: any epoxy uncured, motor not fully seated, avionics LED solid or absent.

STRUCTURES DONE

Section 2

Avionics

Firmware, preflight, GO/NO-GO, recovery cut-out

2.1 T-1

Firmware deploy (Dash's laptop, board on USB)

```
cd ~/Documents/mpr-altitude-logger
git status          # clean tree, or know what's dirty
pnpm version        # must print: Firmware version: 2.0.0
pnpm deploy:pico    # wait for VERIFY_OK at the bottom
```

- ☐ If any module reports FAIL, STOP — fix on the bench, do not bring a half-deployed board to the field

Bench preflight (smoke test on home network/desk before packing)

- ☐ pnpm preflight
- ☐ Confirm all rails green, I2C 0x77 present, BMP180 chip ID OK, RAM free > 50 KB
- ☐ Press [Q] to quit cleanly

10-minute logging stress test (T-1 only — too slow for launch day)

The board flies for less than a minute, but a degraded SD or marginal timing only shows up over many minutes. Run this once the night before.

- ☐ pnpm preflight → [R] recalibrate → [B] boot to PAD
- ☐ Disconnect USB. Power the board from a 9V (or leave on USB if you trust the host won't sleep) for at least 10 minutes
- ☐ LED should be slow-blink (PAD) the whole time. If it ever goes solid, abort the test and investigate.
- ☐ After 10 min, power off, pull SD, insert into laptop, then:

```
pnpm decode -- "/Volumes/AVIONICS/AVIONICS_flight_NNN.bin" --plot
```

- ☐ Verify, at 50 Hz sample rate:
 - ☐ Frame count $\approx 30,000 \pm 1\%$ over 600 s (i.e. 29,700–30,300 frames)
 - ☐ Mean sample interval = 20 ms $\pm$ 1 ms
 - ☐ No timing gap > 50 ms (one missed frame is suspicious; multiple = bad)
 - ☐ No [CRASH REBOOT] line in the boot log on next power-up
 - ☐ File grew monotonically (no truncation), no \xAA\x55 resync gaps in the middle
- ☐ If frame count is short, intervals are off, or there are gaps, do NOT fly — swap SD card and re-run. If a fresh SD also fails, investigate firmware before bringing the board to the field.

SD card prep

- ☐ Confirm SD seated in the avionics PCB
- ☐ FAT32 formatted, > 50 MB free (check the existing **flights/** folder isn't full)
- ☐ Spare SD card formatted FAT32 and packed

Battery prep

- ☐ 2× fresh 9V alkaline (one flight, one spare)
- ☐ Unloaded multimeter check — reject anything under 9.0 V
- ☐ Tape over terminals until install

2.2 T-0 (Launch Day)

Bench preflight at the field (do this BEFORE Aerostructures starts nosecone integration)

- ☐ Connect Pico to laptop via USB. Wait for `/dev/cu.usbmodem*` to appear.

```
cd ~/Documents/mpr-altitude-logger
pnpm preflight
```

- ☐ Verify EVERY line is green:
 - ☐ I2C scan finds device 0x77
 - ☐ BMP180 chip ID OK
 - ☐ Pressure within ± 5 hPa of weather report (sanity-check the sensor)
 - ☐ 3V3 rail in 3.0–3.6 V
 - ☐ 5V rail in 4.5–5.5 V
 - ☐ 9V rail in 8.0–10.0 V — if low, swap battery now
 - ☐ SD mounted, free space > 50 MB
 - ☐ Frame size = 40 bytes (v3 log format)
- ☐ If anything is red, run `pnpm pico:diag` for raw live diagnostics

Calibrate and arm

- ☐ Press [R] to recalibrate ground pressure at pad elevation
- ☐ Watch the live altitude trace for 10–15 seconds standing still
- ☐ Filtered altitude must hold within ± 1 m and not drift. If it drifts, recalibrate. If it still drifts, abort to NO-GO.
- ☐ Press [B] to boot into flight mode
- ☐ TUI exits cleanly. LED transitions to slow blink (1 s on / 1 s off) = PAD state, armed, logging-ready
- ☐ Disconnect USB. Do NOT pull power yet — flight battery only from here on

Hand off to Aerostructures

- ☐ Coordinate the “Add battery” → “Install flight computer” → “Install lower bulkhead” sequence
- ☐ Battery should be installed within 1 hour of expected launch slot — phone timer mandatory
- ☐ After battery is connected and PCB installed, confirm slow-blink LED is visible through the nosecone vent before Aerostructures closes the bulkhead

Final pad check (just before vertical)

- ☐ Visually confirm slow-blink LED through the pressure hole / vent
- ☐ Slow blink only = GO. Anything else = NO-GO
- ☐ No laptop interaction at the pad. The board self-detects launch (alt > 15 m AND velocity > 10 m/s, sustained 0.5 s)

LED reference (memorise this)

- Fast blink (250 ms) — booting / preflight in progress
- Solid ON — error
- **Slow blink (1 s) — PAD, ready, safe to fly**
- Fast blink (50 ms) — BOOST detected
- Medium blink — COAST / DROGUE / MAIN descent
- Double flash — APOGEE
- Triple flash — LANDED, data safe

2.3 Post-flight (after recovery)

The avionics PCB is held by 3× M3 screws, but the lower bulkhead is epoxied in place — so the bay is sealed and the screws cannot be reached without cutting the nosecone open. Power is still live until the battery is unplugged — treat the board as energised.

Cut the nosecone

Choose ONE cut location:

- ☐ **Option A** — circumferential cut *just below the avionics* (between PCB and the lower bulkhead). Exposes the battery side first.
- ☐ **Option B** — circumferential cut *just above the avionics* (between PCB and the nose tip / ballast). Exposes the PCB top side first.
- ☐ Mark the cut line with a pencil all the way around before cutting
- ☐ Use a hand saw, rotary tool, or sharp knife — slow and steady, multiple shallow passes
- ☐ Pause and inspect every few mm. Pull debris out as you go.

WARNING

Stop cutting the moment you feel the wall give. Do NOT push the blade past the wall thickness — the board, battery, or wiring is directly behind it.

Disconnect the battery FIRST

- ☐ Once the cut is open, locate the battery clip and disconnect it before doing anything else
- ☐ LED should go dark — confirms power is off
- ☐ Tape over the 9V terminals immediately

Free the PCB and SD

- ☐ Brush or blow swarf out of the bay before unscrewing
- ☐ Reach through the cut and unscrew the 3× M3 PCB screws
- ☐ Catch each screw as it comes out — if one drops into the bay it can roll under the PCB and short it. Account for all 3.
- ☐ Lift the PCB out cleanly, supporting it by the edges (not by components)
- ☐ Pull the SD card out cleanly

WARNING

Do NOT let a screw, screwdriver tip, blade, or piece of plastic swarf touch the board while power is live or after. Debris bridging pads can short the board and brick it.

Decode the flight

- ☐ Insert the SD card into laptop — auto-mounts as `/Volumes/AVIONICS/`
- ☐ `pnpm postflight` — TUI auto-detects SD, lists flights newest first, decodes selected flight

- ☐ Save the `.bin`, `.csv`, and `_report.txt` to the `flights/` folder
- ☐ If summary looks anomalous: `pnpm decode - "flights/AVIONICS_flight_NNN.bin" -plot` to inspect raw frames
- ☐ If a `[CRASH REBOOT]` line appears in the boot log, save the file and flag it before the next flight — do not reuse the SD until investigated

Multi-flight day — between flights

- ☐ Disconnect avionics power, eject and reseal SD, swap or top up 9V
- ☐ USB → laptop → `pnpm preflight` → [R] recalibrate → [B] boot
- ☐ Disconnect → into airframe
- ☐ Each flight gets an auto-incremented filename (`AVIONICS_flight_NNN.bin`) — no overwrite risk

GO / NO-GO

GO: all preflight rails green, ground pressure calibrated, slow-blink LED on PAD, SD mounted with > 50 MB free.

NO-GO (any of these = scratch the flight):

- ☐ Solid LED at any point after preflight
- ☐ 9V rail < 8.5 V on TUI
- ☐ Altitude drift > 2 m on bench
- ☐ SD won't mount or < 10 MB free
- ☐ Frame timing > 30 ms in `pico_diag`
- ☐ Stuck fast-blink LED that won't settle to slow blink within 30 s of boot

Emergency: avionics won't boot at the pad

1. Solid LED → known-bad state. No spare PCB — scratch the flight.
2. No LED at all → power. Check 9V, switch, connector.
3. Stuck fast-blink → power cycle once. If it repeats, swap SD card first (cheapest, fastest fix).

When in doubt, scratch. A scratched flight is recoverable; a corrupt log is not.

AVIONICS DONE!

Section 3

Recovery

Chute pack, field recovery, avionics handling

Deployment: chute is deployed by the motor's delay charge. There are no separate pyro charges, e-matches, or electronic deployment. The motor's delay grain ejects the chute via the forward closure when the propellant burns through.

3.1 T-1

Chute inspection

- ☐ Unfold chute fully, lay flat
- ☐ Check canopy for tears, scorch marks, melted shroud lines
- ☐ Check shock cord for fraying, knots holding firm
- ☐ Check swivels and quick links for damage
- ☐ Repack chute loosely (final pack happens T-0)

3.2 T-0 (Launch Day)

Chute pack and install

- ☐ Fold chute per team's standard fold (Z-fold or roll, whatever you train on)
- ☐ Wrap in nomex or chute protector to shield from motor exhaust gases
- ☐ Connect to shock cord, shock cord to eyebolt on lower bulkhead
- ☐ Pack into airframe, leaving room above for ejection
- ☐ Confirm motor delay matches predicted apogee time from ORK sim — if delay is too short the chute deploys under high speed and shreds; too long and the rocket lawn-darts

WARNING

The motor delay must match the predicted coast time. Verify the motor designation includes the right delay (e.g. H100-7 = 7-second delay). A wrong delay = destroyed rocket. Check twice before motor install.

GO / NO-GO

GO: chute inspected, packed, shock cord connected, motor delay matches sim (± 1 s).

NO-GO: damaged canopy, frayed shock cord, motor delay mismatch.

3.3 Field Recovery (post-launch)

Watch and track

- ☐ Eyes on the rocket from launch to landing — assign a designated tracker
- ☐ Note apogee location and chute deployment (visible puff)
- ☐ Track descent direction; note landmarks near landing zone
- ☐ Wait for Range Director's clearance before walking onto the field

Walk-out

- ☐ Recovery team: 2+ people, hi-vis vests, phone with GPS, radio if available
- ☐ Don't cross active landing zones for other rockets until cleared
- ☐ Photograph the landing site before touching — captures orientation and any damage

Approach the rocket

- ☐ Spent motor case may still be hot — give it a few minutes if you arrive fast
- ☐ Inspect for damage: bent fins, cracked airframe, tangled chute, missing nosecone
- ☐ Check chute condition for post-flight notes

Avionics handling in the field

The avionics is still powered. The board's last act is to detect LANDED and flush the SD — the LED tells you whether that worked.

- ☐ Look at the LED through the nosecone vent / pressure hole:
 - ☐ **Triple flash** = LANDED, data flushed, safe to transport. Best case.
 - ☐ **Slow blink** = still in PAD — launch was never detected, log will be short or empty
 - ☐ **Other pattern** = stuck in a mid-flight state, may not have flushed cleanly
- ☐ DO NOT disconnect the battery in the field unless you have to. Let the board sit at LANDED so any pending writes finish.
- ☐ If the LED is in an odd state, photograph it before moving — useful for post-flight diagnosis

WARNING

Vibration during SD writes can corrupt the last few frames. If the board is still mid-flush (slow LED, not yet triple flash), avoid jostling the rocket. Walking it back is fine; throwing it in the back of a car is not.

Transport back to bench

- ☐ Carry the rocket supported, not by the nosecone (cumulative shock isn't great for the SD)
- ☐ Battery stays connected. If transport is > 30 min and the LED is showing triple flash (LANDED), you can power off if needed — the data is safe.

- ☐ Bag the rocket if it's wet, muddy, or going through dust
- ☐ Once at the bench, hand off to Avionics for the post-flight cut-out (see Avionics section)

If the rocket is lost / unrecoverable

- ☐ Mark the last seen position on a map (phone GPS pin)
- ☐ Search in expanding circles from predicted landing zone
- ☐ Ask other teams if they spotted it
- ☐ If unrecoverable: the SD card and battery are in the rocket, and the data is gone. Note for post-mortem.

RECOVERY DONE

Section 4

Propulsions

Motor, igniter, paperwork

TO FILL IN — owner: _____

Suggested structure:

- **T-1:** motor inspection, igniter inventory, certification paperwork
- **T-0:** motor install (coordinate with Aerostructures), igniter install, continuity check, GO/NO-GO criteria
- **Post-flight:** motor case retrieval, post-fire inspection

Section 5

Master Timeline (T-0)

Sequencing and comms

Approximate, adjust to launch slot:

Time	Owner	Action
Slot - 3:00 h	All	Arrive, unpack, set up bench
Slot - 2:30 h	Aerostructures	Eyebolt, motor friction fit, ballast prep
Slot - 2:00 h	Avionics	Bench preflight, calibrate, boot to PAD
Slot - 1:30 h	Avionics + Aerostructures	Battery install, PCB install in nosecone
Slot - 1:15 h	Aerostructures	Lower bulkhead epoxy, nosecone integration
Slot - 0:45 h	Recovery + Propulsions	Charges, igniter, final continuity
Slot - 0:30 h	All	Walk to pad, vertical, final visual checks
Slot - 0:10 h	Range	Range crew takeover
Slot 0:00	—	LAUNCH
Slot + 0:30 h	All	Recovery walk
Slot + 1:00 h	Avionics	Cut nosecone, disconnect battery, pull SD, post-flight decode

5.1 Comms

- ☐ **Launch Director** calls GO/NO-GO at each phase. Anyone can call NO-GO at any time for any reason.
- ☐ Subsystem leads must verbally confirm GO at handoff points (Avionics → Aerostructures battery install, etc.)

Section 6

Pack List

What goes in the box

6.1 Structures

- ☐ Spare nosecone
- ☐ 5-minute epoxy (lower bulkhead)
- ☐ 3× M3 magnetic screws (avionics PCB) + spares
- ☐ Magnetic screwdriver (M3 bit) — PCB install + post-flight removal
- ☐ Split shot sinkers
- ☐ Eye bolt
- ☐ Drill + drill bits (eyebolt + pressure hole)
- ☐ Toothpicks (epoxy cure check)
- ☐ Electrical tape, hot glue gun + sticks (fallback)
- ☐ Hand saw / Dremel / sharp knife (for post-flight nosecone cut)
- ☐ Bench vice or clamp (to hold nosecone steady during cut)
- ☐ Safety glasses

6.2 Avionics

- ☐ Laptop + charger (Dash)
- ☐ USB-A to micro-USB cable — DATA capable, not charge-only
- ☐ 2× fresh 9V batteries (alkaline)
- ☐ Multimeter
- ☐ Spare SD card, formatted FAT32
- ☐ USB SD card reader
- ☐ Printout of this doc (no field wifi guaranteed)

6.3 Recovery

- ☐ Parachute (main + drogue if applicable)
- ☐ Shock cord, swivels
- ☐ Ejection charges (per manifest)
- ☐ Igniter / e-match wire
- ☐ Continuity tester
- ☐ Nomex / chute protector
- ☐ Spare quick links

6.4 Propulsions

- ☐ Motor (per manifest)
- ☐ Spare igniter(s)
- ☐ Motor retention clip
- ☐ Cert paperwork
- ☐ Igniter wire
- ☐ Continuity tester (shared with Recovery OK)

6.5 Shared

- ☐ First aid kit
- ☐ Hi-vis vests
- ☐ Sunscreen, water, snacks
- ☐ Folding table, chairs
- ☐ Power bank for laptop